

Dot Product, Cosine, and Euclidean Distance Are Three Different Questions

Why the choice of similarity metric — not the embedding model — often decides what "similar" means in a retrieval system.

I used to think cosine similarity was just the correct answer for vector search. Then I realized I had memorized the metric before understanding the question.

Dot product, cosine similarity, and Euclidean distance are not three versions of the same idea. They are three different questions — and depending on which one is chosen, the word *similar* means something completely different.

A minimal working example

Take two vectors that point in exactly the same direction — $\mathbf{y}$ is simply $\mathbf{x}$ scaled down by five.

VECTOR X

[3, 4]

VECTOR Y

[0.6, 0.8]

Same direction, different length — now ask all three questions of the same pair.

DOT PRODUCT

5.0

magnitude still counts

COSINE SIMILARITY

1.0

perfectly aligned

EUCLIDEAN DISTANCE

4.0

far apart in space

Same two vectors. Three different answers. The dot product returns **5.0** because magnitude is still part of the computation. Cosine sees perfect similarity. Euclidean sees two points sitting far apart in space.

1 The dot product keeps magnitude alive

DEFINITION

$$\mathbf{x} \cdot \mathbf{y} = \sum_i x_i y_i = \|\mathbf{x}\| \|\mathbf{y}\| \cos \theta$$

Both the angle and the sizes live inside that single expression. Scale a vector up and the dot product is amplified. In Transformer attention that behavior is intentional: larger dot products produce larger attention scores before the softmax. The model is not just asking which tokens point in similar directions — it is asking which ones are strong enough to matter.

2 Cosine similarity divides magnitude out

DEFINITION

$$\cos(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

The division happens inside the formula itself. Direction is what survives it. $\mathbf{x}$ and $\mathbf{y}$ score 1.0 not because they are identical, but because they point the same way — length does not dominate the score. Direction does.

3 Euclidean distance asks something neither of them asks

DEFINITION

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2 = \sqrt{\sum_i (x_i - y_i)^2}$$

$\mathbf{x}$ and $\mathbf{y}$ are 4.0 units apart in space. Same direction, very different positions — that gap is real, and Euclidean distance captures it honestly. It is the only one of the three questions that cares about absolute position rather than orientation or amplified alignment.

When the three questions collapse into one

Now normalize $\mathbf{x}$ to unit length:

UNIT VECTOR

$$\hat{\mathbf{x}} = [0.6, 0.8]$$

$\hat{\mathbf{x}}$ and $\mathbf{y}$ are now identical. And on unit-normalized vectors, two clean identities hold:

IDENTITY ONE

$$\mathbf{x} \cdot \mathbf{y} = \cos(\mathbf{x}, \mathbf{y})$$

IDENTITY TWO

$$\|\mathbf{x} - \mathbf{y}\|_2^2 = 2 - 2\cos(\mathbf{x}, \mathbf{y})$$

WHY THIS MATTERS

Maximizing the dot product, maximizing cosine similarity, and minimizing Euclidean distance now produce the **same ranking**. The three questions collapse into one answer — but only once normalization has removed magnitude from the computation.

Normalization is not just preprocessing. It is a decision about whether magnitude is allowed to participate in the computation at all. Skip that decision, and the choice of similarity measure silently decides what "nearest neighbor" means.

TAKEAWAY

Sometimes the embeddings are fine. The index is fine. The question was never defined.

Choosing between the three, in practice

As a working rule: reach for the **dot product** when magnitude itself is meaningful — attention scores, popularity-weighted retrieval, learned scoring functions. Reach for **cosine similarity** when only direction or semantic orientation should count, which is why it dominates embedding search. Reach for **Euclidean distance** when absolute closeness in a real coordinate space is the actual question, as in k-means or physical nearest-neighbor search. A one-page visual summary of all three follows on the next page.

Similarity Math for AI Engineers

Dot Product · Cosine Similarity · Euclidean Distance

Dot Product

WHAT IT MEASURES

Alignment weighted by magnitude.

FORMULA

$$\mathbf{x} \cdot \mathbf{y} = \sum x_i y_i = \|\mathbf{x}\| \|\mathbf{y}\| \cos \theta$$

KEY PROPERTIES

- Depends on both angle and vector length
- Not scale-invariant
- Range for real vectors: $(-\infty, +\infty)$
- Scaling a vector amplifies the score

PRACTICAL AI USES

- Attention logits before softmax (scaled dot-product attention)
- Maximum inner-product search (MIPS)
- Scoring when magnitude is meaningful

BEST WHEN

Magnitude carries information.

Cosine Similarity

WHAT IT MEASURES

Directional alignment, independent of vector length.

FORMULA

$$\cos(\mathbf{x}, \mathbf{y}) = (\mathbf{x} \cdot \mathbf{y}) / (\|\mathbf{x}\| \|\mathbf{y}\|)$$

KEY PROPERTIES

- Removes magnitude via normalization
- Invariant to positive rescaling
- Range for real vectors: $[-1, 1]$
- A similarity score, not a metric

PRACTICAL AI USES

- Embedding similarity / semantic search
- Retrieval on normalized embeddings
- Cosine-style clustering (e.g. spherical k-means)

BEST WHEN

Direction matters more than length.

Euclidean Distance

WHAT IT MEASURES

Straight-line distance in space.

FORMULA

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2 = \sqrt{\sum (x_i - y_i)^2}$$

KEY PROPERTIES

- Smaller distance means greater similarity
- Sensitive to scale and magnitude
- Range: $[0, \infty)$
- A proper metric: non-negativity, identity, symmetry, triangle inequality

PRACTICAL AI USES

- k-nearest neighbors
- k-means (minimizes squared Euclidean distance)
- Metric-space nearest-neighbor search

BEST WHEN

Absolute closeness matters.

HOW THEY COMPARE

Dot product asks: how aligned *and* how large are they?

Cosine similarity asks: how aligned are they, regardless of length?

Euclidean distance asks: how far apart are they in space?

IMPORTANT TRUTHS

1. Normalize embeddings when direction should matter more than magnitude.
2. On unit-normalized vectors: $\mathbf{x} \cdot \mathbf{y} = \cos(\mathbf{x}, \mathbf{y})$.
3. On unit-normalized vectors: $\|\mathbf{x} - \mathbf{y}\|_2^2 = 2 - 2\cos(\mathbf{x}, \mathbf{y})$.
4. On unit-normalized vectors, maximizing dot product / cosine and minimizing Euclidean distance give the **same** nearest-neighbor ranking.
5. Without normalization, the choice of measure can change retrieval behavior entirely.

METRIC NOTE

Euclidean distance is a proper metric. **Cosine similarity is not a metric** — and the common cosine distance, $1 - \cos(\mathbf{x}, \mathbf{y})$, is not a proper metric either.

Angular distance, $\arccos(\cos(\mathbf{x}, \mathbf{y}))$, is the metric version of cosine similarity on unit-normalized vectors.

KEY TAKEAWAY



Use cosine similarity when direction matters, dot product when magnitude should influence the score, and Euclidean distance when absolute closeness in space is the actual question.